Juhwan Lee

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SUMMARY

Ph.D. student and research assistant in Air Transportation at Korea Aerospace University, with experience in airport operations, air traffic management, aviation simulation, capacity and delay analysis, and data-driven forecasting. My research focuses on understanding dynamic aircraft and passenger flows and integrating simulation and AI-based prediction into decision support systems to enable proactive, adaptive, and efficient airport operations.

EDUCATION

Korea Aerospace University (KAU), Goyang, South Korea *Sep 2023 – Present*
Ph.D. Student in Air Transportation, Transportation and Logistics
Research focus: airport operations, capacity analysis, passenger terminal simulation, and data-driven queue forecasting

Korea Aerospace University (KAU), Goyang, South Korea *August 2023*
M.S. in Air Transportation, Transportation and Logistics
GPA: 4.5/4.5
Related courses: Airport Design with Simulation, Theory and Applications of Data Mining

Korea Aerospace University (KAU), Goyang, South Korea *August 2021*
B.S. in Air Transportation, Transportation and Logistics
GPA: 4.05/4.5
Related courses: Introductory Applied Machine Learning, Air Traffic Systems Analysis, Airspace Systems Analysis, Computer Applications in Air Traffic Management

RESEARCH INTERESTS

Data-Driven Airport Operations and Aviation Decision Support

- Airport capacity and demand–capacity imbalance analysis
- Passenger terminal simulation and data-driven congestion forecasting
- AI-enabled decision support for airport operations and resource allocation

PUBLICATIONS

[1] **Juhwan Lee**, S. Yoon, K. Lee, S. Jeong, and H. Baik, "Airport Passenger Queue Forecasting for Departure Gates and Security Checkpoints," IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), 2026.

[2] J. Shin, H. Baik, **Juhwan Lee**, and B. Yi, "Network Connectivity Analysis of Major Asian Airports Using the NetScan Model:Focusing on Incheon Internation Airport," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[3] J. Shin, B. Yi, M. Kim, J. Cho, and **Juhwan Lee**, "Study on Airport Capacity Utilization Analysis Using the Capacity Coverage Chart (CCC)," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[4] E. Kim, **Juhwan Lee**, Y. Shin, and H. Baik, "A Study on Airport Access Mode Choice Behavior Using Mobile Data: Case of Incheon International Airport," IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), 2023.

[5] Y. Lee, **Juhwan Lee**, and K. Lee, "A Study on the Visualization of an Airline's Fleet State Variation," *Journal of the Korean Society for Aviation and Aeronautics*, 2021.

PROJECT EXPERIENCE

Integrated Information System Enhancement Project
(Sep 2025 – Mar 2026)

- Analyzed airport operational data, including flight schedules, passenger flows, and observed queue lengths and waiting times.
- Developed and validated a CAST-based passenger terminal simulation model.

Strategic Planning for Enhancing Hub Competitiveness of Incheon International Airport
(Oct 2025 – Dec 2025) [2],[3]

- Analyzed changes in delay according to passenger demand and capacity conditions.
- Supported strategic evaluation of airport competitiveness from an operational performance perspective.

Development of Passenger Terminal Simulation Models for Incheon International Airport
(Jul 2025 – Dec 2025)

- Developed MATLAB-based simulation models for departure passenger processing facilities using actual airport observation data.
- Estimated queue length and waiting time under time-varying demand and operational conditions.

Analysis of Variable Multiple Taxi-Line Operations for Apron Congestion Mitigation at Jeju International Airport
(Jul 2025 – Dec 2025)

- Simulated apron congestion and aircraft surface movement using TAAM.
- Evaluated taxi-line operational alternatives for congestion mitigation and surface movement efficiency.

Master Plan for Incheon International Airport
(Sep 2022 – Aug 2024) [4]

- Performed runway capacity analysis using TAAM and passenger terminal simulation using CAST.
- Evaluated capacity, congestion, delay, and economic feasibility for airport master planning.

Performance-Based Navigation (PBN) Flight Procedure Maintenance Service
(Sep 2021 – Dec 2021)

- Analyzed observed aircraft trajectory data to evaluate the effectiveness of flight procedures under the operation of Continuous Descent Operations (CDO) and Continuous Climb Operations (CCO).

AWARDS AND ACHIEVEMENTS

Scholarship, KAU Professional Talent Development Scholarship	$4,520	*Fall 2021*
2nd Place in Capstone Design Competition	$260	*Fall 2020*
2nd Place in ICT Startup Makerthon	$740	*Fall 2019*
Scholarship, Jeongseok C	$841	*Fall 2019*
Scholarship, Jeongseok C	$739	*Fall 2016*

TECHNICAL SKILLS

Programming	Python, MATLAB, LaTeX
AI / Data Analysis	PyTorch, Time-series forecasting, Transformer-based modeling, Statistical analysis
Simulation	TAAM for runway and surface operation simulation; CAST and MATLAB-based passenger terminal simulation
Aviation Analysis	Airport capacity and delay analysis, Queue and waiting time analysis